

Chapter 8.2

Reviews

Continuing...

- **Peer reviews (Inspections and Walkthroughs)**
 - **Participants**
 - **Preparations**
 - **The FDR session**
 - **Post-review activities**
 - **Peer review coverage**
- **Comparison of peer reviews methods**
- **Expert opinions**

Peer Reviews

- Will discuss
 - 1. inspections and
 - 2. walkthroughs.
- Difference between formal design reviews and peer reviews is really in both their participants and authority.
- DRs: most participants **hold superior positions** to the project leaders and customer reps;
- Peer reviews, we have equals
 - members of his/her department and other units.

Peer Reviews

- Other major difference is
 - degree of authority and
 - objective of each review method.
- FDRs: authorized to approved design doc
 - work can now continue in project.
- Not granted in peer reviews
 - main objectives lie in
 - detecting errors and
 - deviations from standards.

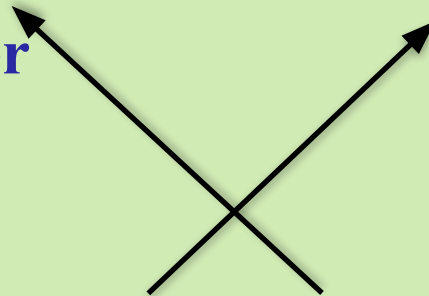
Participants of Peer Reviews

Inspection

- Review leader
- The author
- Specialized professionals:
 - Designer
 - Coder or implementer
 - Tester

Walkthrough

- Review leader
- The author
- Specialized professionals:
 - Standards enforcer
 - Maintenance expert
 - User representative



Peer Reviews - more

- Tendency nowadays: diminish the value of manual reviews such as inspections and reviews.
- We don't have as many as we used to.
- Largely due to agile approaches to development and concerns for time savings and streamlining the process wherever possible.
- **Empirical evidence**, however, indicates **convincing evidence** that peer reviews are highly efficient and effective.

Peer Reviews:

Inspections / Walk-Throughs

- Walkthroughs and inspection differ in formality –
- Inspections emphasize the **objective of corrective action; more formal**
- Walkthroughs limited to **comments** on document reviewed.
- Inspections also look to **improve methods** as well.
- Inspections are considered **to contribute more** to general level of SQA.

Peer Reviews: Inspections

- Inspections usually based on a comprehensive infrastructure:
 - Development of **inspection checklists** for each type of design document as well as coding languages, which are periodically updated.
 - Development of **typical defect type frequency tables**, based on past findings to direct inspectors to potential ‘defect concentration areas.’
 - **Training of competent professionals** in inspection process issues – making it possible for them to serve as inspection leaders (moderators) or inspection team members
 - Periodic **analysis of the effectiveness of past inspections** to improve the inspection methodology
 - **Introduction of scheduled inspections** into project activity plan and allocation of the required resources, **including resources** for corrections

Comparing the Two...

- Organizations typically modify methods to local considerations.
- Local protocols, team structure, etc. can be modified.
- So, differences between the two can be easily blurred.
- Some view one as the other; vice versa
- Some argue that one is better than the other, and vice versa.
- Research has indicated that walkthroughs discover **far fewer** defects found but at the **same cost**.
- Likely this is due to some of the lost formalisms??

PARTICIPANTS

Inspection

Moderator (scribe)

Author

Coder or
implementer
(presenter)

Designer

Tester

Walkthrough

Coordinator (scribe)

Maintenance
expertStandards
enforcerAuthor
(presenter)User
representative

PROCESS

Organizational
preparations

Overview meeting

Thorough review of
document

Inspection session(s)

Inspection session report
Inspection summary reportCorrections
and reworkingFollow-up of corrections
and reworkingOrganizational
preparationsBrief overview
reading

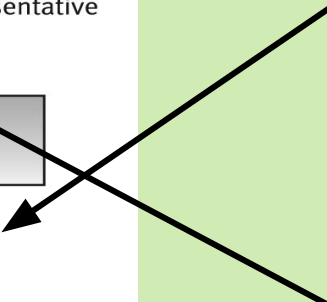
Walkthrough session(s)

Walkthrough
session report

Note: author is
not the presenter



Much less
formal...



Author **is** the
presenter

Focus on Peer Reviews:

- So for these two peer review methods, we will look at:
 - Participants of peer reviews
 - Preparation for peer reviews (some major differences)
 - The peer review session (presenters and emphases are different)
 - Post peer-review activities (differ considerably)
 - Peer review efficiency (arguable)
- The principles that we talk about can apply to both **design** peer reviews and **code** peer reviews.

Participants of Peer Reviews

- Optimally, 3-4 participants
- Should be **peers** of the software system designer-author.
 - This allows for free discussion without any intimidation.
- Need a **good blend** of individual types: a review leader, the author, and specialized professionals as needed for the focus of the review.
- **Review Leader**
 - Moderator in inspections; Coordinator in walkthroughs.
 - Must be well-versed in project development and current technologies
 - Have good relationships with author and development team
 - Come from **outside** the project team
 - History of **proven experience** in coordination / leadership settings like this.
 - For inspections, **training** as a moderator is required.

Participants of Peer Reviews

- **Specialized Professionals** – note these are experienced folks.
- These differ by review type: **inspections** / walkthroughs
- **Inspections**: (Walk-throughs - next slide)
 - **A designer** – generally the **systems analyst** responsible for analysis and design of software system reviewed
 - **A coder or implementer** – one who is **thoroughly acquainted** with coding tasks, preferably the **leader** of the designated coding team.
 - Able to detect defects leading to coding errors and other **software implementation** issues.
 - **A tester** – experienced professional, **preferably leader of assigned testing team** who focuses on identification of design errors usually detected during the testing phase.

Participants of Peer Reviews

- Specialized Professionals differ by review type : **walkthroughs**
- **Walk Throughs:**
 - **A standards enforcer** – team member specialized in development standards and procedures;
 - locate deviations from these standards and procedures. (coding standards, such as indentation, data naming, program organization, such as class content (coupling, cohesion....)
 - These problems substantially affect the team's long-term effectiveness for both **development and follow-on maintenance**.
 - **A maintenance expert** – focus on maintainability / testability issues
 - to detect design defects that may hinder bug correction and impact performance of future changes. Example: refactoring expert;
 - **A maintenance expert** - Focuses also on documentation (completeness / correctness) vital for maintenance activity.
 - **A user representation** – need an internal user (if customer is in the unit) or an external representative - review's validity due to his/her point of view as user-customer rather than the designer-supplier.

Participants of Peer Reviews

- **Team Assignments**

- **Presenter:**

- For **inspections:**

- The presenter of document; chosen by the moderator; should **not be** document's author
 - Sometimes the software **coder serves as presenter** due to the familiarity of the design logic and its implications for coding.

- For **walk-throughs:**

- Author most familiar with the document should be chosen to present it to the group.
 - Could be design document developer; programmer for code walk-through....
 - Some argue that a neutral person should be used..

- **Scribe:**

- Team leader will often serve as the scribe and record noted defects to be corrected.

16 Preparation for a Peer Review Session

- Peer Review **Leader's** Preparation for the session:
- **For Design Document:**
 - Select the most difficult / complex sections; sections prone to defects.
 - The most critical sections / where defect can cause severe damage
- **Select team members**
 - Limit review session to two hours – absolutely.
 - Schedule up to two sessions a day if review tasks is sizable.
 - Schedule right after the document is ready for inspection. Don't wait....
- **Distribute the document** to the team members prior to the review session.

17 Preparation for a Peer Review Session

- **Peer review team's** preparations for review session:
- **For inspections:** team members, preparation is quite thorough;
 - For **walkthrough** brief.
 - **Inspection:** participants **must read document** and **list comments before** inspection begins.
 - In **overview meeting**, the author provides inspection team members with the necessary background for reviewing chosen document, project in general, logic, processes, outputs, inputs, interfaces.
 - **Tool** for inspector's review: **checklist** for specific documents.
- **For walkthroughs:** team briefly reads materials for general overview of project
 - Generally they lack detailed knowledge and its substantive area.
 - In most cases, **team participants not required to prepare advance comments.**

The Peer Review Session

- Procedurally, presenter reads document section and may add an explanation.
- Participants may offer comments on doc or on comments
- **Restrict discussion** to identification of errors – **no solutions.**
- **Presenter** in walkthrough provides an overview
- **Walkthrough Scribe** records each error (location, description, type) – incorrect, missing, etc.
- **Inspection scribe** will add estimated severity of each defect, a factor to be used in the statistical analysis of defects found and for the foundation of preventive / corrective actions.

The Peer Review Session

- See table 8.1: classifies errors from 5 to 1 (major to minor)
- **Session Documentation**
 - **For inspections** – much more comprehensive
 - Inspection Session **Findings** Report – produced by scribe
 - Inspection Session **Summary** Report – compiled by inspection leader after session or series of sessions dealing with the same document
 - Report summarizes inspection findings and resources invested in the inspections...
 - Report serves as inputs for analysis aimed at inspection process improvement and corrective actions that go beyond the specific document or project.
 - **For walkthroughs** – copies of the error documentation should be provided to the development team and the session participants.

The Post Review Session

- Here is the most fundamental differentiating element between the two peer review methods.
- **Inspection:**
 - **Does not end** with a review session or distribution of reports
 - **Post inspection activities** are conducted to attest to:
 - Prompt, effective correction / reworking of all errors
 - Transmission of the inspection reports to **controlling authority** for analysis

The Efficiency of Peer Reviews

- These activities are under constant debate.
- Some of the **more common metrics** applied to estimate the effectiveness of peer reviews, suggested by literature:
 - **Peer review detection efficiency** (average hours worked per defect detected)
 - **Peer review defect detection density** (average number of defects detected per page of the design document)
 - **Internal peer review effectiveness** (percentage of defects detected by peer review as a percentage of total defects detected by the developer)

The Efficiency of Peer Reviews

- (Not a lot of data on findings) An interesting study undertaken by Cusumano reports results of a study on the effectiveness of design review, code inspection, and testing at Fujitsu from 1977 to 1982.
- Findings are still of interest, **as data shows substantial improvement in software quality associated with an increased share of code inspection and design reviews and a reduced share of software testing.**
- Software quality measured here by the number of defects per 1000 lines of maintained code, detected by the users during the first six months of regular software system use.
- The results only refer to the inspection method; one **guesses** a similar result would apply to walkthrough methods.

Code inspection effectiveness at Fujitso (Cusumano)

Year	Defect detection method			Defects per 1000 lines of maintained code
	Test %	Design review %	Code inspection %	
1977	85	---	15	0.19
1978	80	5	15	0.13
1979	70	10	20	0.06
1980	60	15	25	0.05
1981	40	30	30	0.04
1982	30	40	30	0.02

Comparison of Team Review Methods

- Consider the table on the next slide that provide a look-back on what is contained / omitted from peer reviews.

Sections recommended to be included in or omitted from peer reviews

Sections recommended for inclusion

- **Sections of complicated logic**
- **Critical sections, where defects severely damage essential system capability**
- **Sections dealing with new environments**
- **Sections designed by new or inexperienced team members**

Sections recommended for omission

- **“Straightforward” sections (no complications)**
- **Sections of a type already reviewed by the team in similar past projects**
- **Sections that, if faulty, are not expected to effect functionality**
- **Reused design and code**
- **Repeated parts of the design and code**

Similarly a review of methodologies:

- Really shows the differences in tabular form.

Comparison of review methodologies - Process of review

Properties	Design review	Inspection	Walkthrough
Overview meeting	No	Yes	No
Participant's preparations	Yes - thorough	Yes - thorough	Yes - brief
Review session	Yes	Yes	Yes
Follow-up of corrections	Yes	Yes	No
Formal training of participants	No	Yes	No
Participant's use of checklists	No	Yes	No
Error-related data collection	Not formally required	Formally required	Not formally required
Review documentation	Formal design review report	1) Inspection session findings report 2) Inspection session summary report	

Expert Opinions

- Most experts support quality evaluation by introducing additional capabilities for the internal review staff.
- The organization's internal quality assurance activities are thereby reinforced.
- Experts suggest participation of an expert or his/her participation as an external review member if the following circumstances apply:

Situations beneficial for expert's participation in reviews

- **Insufficient in-house professional capabilities in a specialized area.**
- **Temporary lack of in-house professionals for review team.**
- **Indecisiveness caused by major disagreements among the organization's senior professionals.**
- **In small organizations, where the number of suitable candidates for a review team is insufficient.**

Homework – Chapter 8

You are to answer questions 8.4 and 8.6

Submit your thoughtful answers via Blackboard Assignments for Chapter 8, as usual.

Team Discussion Questions

1. Briefly discuss the main concepts in Chapter 8 emphasizing the topics that I did not finish in lecture. You may use my slides as appropriate.
2. Discuss question 8.2
3. Discuss question 8.3